



Information Technology Institute

Docker and Kubernetes

homework project

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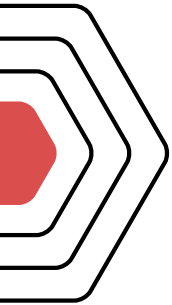
2026

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HOMEWORK N°



Main project



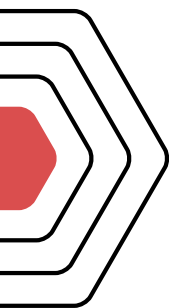
Cloning the project repo

```

islām-tarek-moawad@lala:~$ git clone https://github.com/Eslam-Fawzi/k8s-demo.git
Cloning into 'k8s-demo'...
remote: Enumerating objects: 32, done.
remote: Counting objects: 100% (32/32), done.
remote: Compressing objects: 100% (28/28), done.
remote: Total 32 (delta 2), reused 32 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (32/32), 11.19 KiB | 5.59 MiB/s, done.
Resolving deltas: 100% (2/2), done.
islām-tarek-moawad@lala:~$ cd k8s-demo
islām-tarek-moawad@lala:~/k8s-demo$ ls
LICENSE  README.md  backend  docker-compose.yml  frontend  k8s  migrator
islām-tarek-moawad@lala:~/k8s-demo$

```

Figure 1.1



Building the images

```
islām-tarek-moawad@lala:~$ cd k8s-demo
islām-tarek-moawad@lala:~/k8s-demo$ cd backend
islām-tarek-moawad@lala:~/k8s-demo/backend$ docker build -t islām2005311/k8s-backend:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 6.144kB
Step 1/7 : FROM python:3.11-slim
--> 5d7182676c84
Step 2/7 : WORKDIR /app
--> Using cache
--> a6662aa74ba1
Step 3/7 : COPY requirements.txt .
--> Using cache
--> d99d684f0d59
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> c418b8915cbb
Step 5/7 : COPY main.py .
--> Using cache
--> 30607fe8a6c3
Step 6/7 : EXPOSE 8000
--> Using cache
--> 3c0fdca686a5
Step 7/7 : CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
--> Using cache
```

Figure 1.2

```
islām-tarek-moawad@lala:~/k8s-demo/backend$ cd ../frontend
islām-tarek-moawad@lala:~/k8s-demo/frontend$ docker build -t islām2005311/k8s-frontend:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 12.29kB
Step 1/7 : FROM nginx:alpine
--> 7bc5ba2f958a
Step 2/7 : COPY index.html /usr/share/nginx/html/index.html
--> Using cache
--> 9ad813a95d4f
Step 3/7 : COPY nginx.conf /etc/nginx/conf.d/default.conf
--> Using cache
--> 52fe7799e0f9
Step 4/7 : COPY entrypoint.sh /entrypoint.sh
--> Using cache
--> 28f6a723d920
Step 5/7 : RUN chmod +x /entrypoint.sh
--> Using cache
--> 61b378ccfba
Step 6/7 : EXPOSE 80
--> Using cache
--> f3f1bdf5de52
Step 7/7 : ENTRYPOINT ["/entrypoint.sh"]
--> Using cache
--> 9fc2d20dd0e9
```

Figure 1.3

```
islām-tarek-moawad@lala:~/k8s-demo/frontend$ cd ../migrator
islām-tarek-moawad@lala:~/k8s-demo/migrator$ docker build -t islām2005311/k8s-migrator:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/5 : FROM python:3.11-slim
--> 5d7182676c84
Step 2/5 : WORKDIR /app
--> Using cache
--> a6662aa74ba1
Step 3/5 : RUN pip install --no-cache-dir psycpg2-binary==2.9.9
--> Running in 527dd65a7867
Collecting psycpg2-binary==2.9.9
  Downloading psycpg2_binary-2.9.9-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.4 kB)
  Downloading psycpg2_binary-2.9.9-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.0 MB)
     3.0/3.0 MB 3.3 MB/s eta 0:00:00
Installing collected packages: psycpg2-binary
Successfully installed psycpg2-binary-2.9.9
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

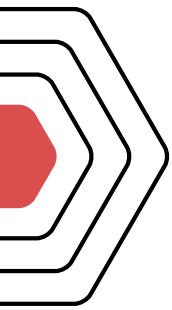
[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
--> Removed intermediate container 527dd65a7867
--> 552a1eb7eb46
```

Figure 1.4

Pushing the docker images

```
islām-tarek-moawad@lala:~/k8s-demo$ docker push islām2005311/k8s-backend:v1
docker push islām2005311/k8s-frontend:v1
docker push islām2005311/k8s-migrator:v1
The push refers to repository [docker.io/islām2005311/k8s-backend]
5de46a4ae8f8: Pushed
18ba56e3aa68: Pushed
36d5d5fd6f2c: Pushed
366aaca33bbb: Pushed
34d8e1a7bf9e: Pushed
99dd0251c41d: Mounted from library/python
3077d73e91e8: Mounted from library/python
411a86676185: Mounted from library/python
v1: digest: sha256:e0bc53f207fb06cbf728936c589f38b931a883af36e4ac2a189306afea859f90 size: 1992
The push refers to repository [docker.io/islām2005311/k8s-frontend]
edae910a238b: Pushed
e0944a9354a9: Pushed
93d06d53857b: Pushed
d622acd40f3e: Pushed
edc23e36f952: Mounted from library/nginx
628c86fdf4c1: Pushed
63c800442e2a: Pushed
31e777702b00: Pushed
e3b86476d58e: Pushed
18ca587a5d44: Pushed
1c4d71cf6bfe: Mounted from library/nginx
34884abbe928: Mounted from library/nginx
v1: digest: sha256:5850bd5f39032ceacfe15f45401c5c891d57e2015eaeca1f5d80c5a207f7808c size: 2818
```

Figure 1.5



Creating a network

```
Thu Aug 27 7:30:57 PM
islam-tarek-moawad@lala: ~/k8s-demo
islam-tarek-moawad@lala:~/k8s-demo$ docker network create k8s-network
4641ba867c46808a5d1ae1455f08291f378d158f2a0fbefbe8cab4ce776bfed3
islam-tarek-moawad@lala:~/k8s-demo$
```

Figure 1.6

Running PostgreSQL

```
islām-tarek-moawad@lala:~/k8s-demo$ docker run -d --name postgres --network k8s-network -e POSTGRES_DB=demodb -e POSTGRES_USER=demo -e POSTGRES_PASSWORD=demo123 postgres:15-alpine
Unable to find image 'postgres:15-alpine' locally
15-alpine: Pulling from library/postgres
55afa1ecc21d: Already exists
2521846a4c4b: Pull complete
516ac23b2d22: Pull complete
d30adf595bd0: Pull complete
a5b616744b9f: Pull complete
ecbc0f97fe87: Pull complete
26065aa66f8d: Pull complete
4989bdaa0ad3: Pull complete
0141deae452e: Pull complete
e7426913c582: Pull complete
236e3b4c2204: Pull complete
Digest: sha256:fe0737ba566a2c5b2a28f34433c0a423261900ec17b9bf7ad115e1aae7e57f1b
Status: Downloaded newer image for postgres:15-alpine
docker: Error response from daemon: Conflict. The container name "/postgres" is already in use by container "f510eff620a833e534891083c9331aa5eb9335f18b71abc526041af3195f00f0". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
islām-tarek-moawad@lala:~/k8s-demo$
```

Figure 1.7

Running the three containers

```
islām-tarek-moawad@lala:~/k8s-demo$ docker run -d --name postgres --network k8s-network -e POSTGRES_DB=demodb -e POSTGRES_USER=demo -e POSTGRES_PASSWORD=demo123 postgres:15-alpine && docker run --rm --network k8s-network -e DB_HOST=postgres -e DB_PORT=5432 -e DB_NAME=demodb -e DB_USER=demo -e DB_PASSWORD=demo123 islam2005311/k8s-migrator:v1 && docker run -d --name backend --network k8s-network -p 8000:8000 -e DB_HOST=postgres -e DB_PORT=5432 -e DB_NAME=demodb -e DB_USER=demo -e DB_PASSWORD=demo123 islam2005311/k8s-backend:v1 && docker run -d --name frontend --network k8s-network -p 3000:80 -e BACKEND_URL=http://localhost:8000 islam2005311/k8s-frontend:v1
docker: Error response from daemon: Conflict. The container name "/postgres" is already in use by container "f510eff620a833e534891083c9331aa5eb9335f18b71abc526041af3195f00f0". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
islām-tarek-moawad@lala:~/k8s-demo$
```

Figure 1.8